S. S. M Jahir Jahan Khan Miru

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SUMMARY

Urban and Regional Planning graduate from BUET with a strong passion for smart mobility, transport analytics, and computer vision. I bring hands-on research and professional experience in developing AI-powered automated vehicle tracking systems, conducting transport microsimulation (SUMO, Vissim), and performing spatial network analysis using advanced GIS tools. Having successfully led over 20 Traffic Impact Assessments (TIAs) at Plansculpt, I am eager to pursue doctoral research focusing on innovative transportation safety, intelligent mobility systems, and data-driven urban infrastructure optimization.

EXPERIENCE

Urban Planner & CEO

PLANSCULPT PRIVATE CONSULTANTS **November 2023 - Present, Dhaka**

- Completed 20+ comprehensive Traffic Impact Assessments (TIAs) utilizing computer vision and predictive modeling for major civil development projects.
- Developed custom AI pipelines (Ultralytics YOLO, OpenCV) to extract vehicle classifications and movement data from video sensors, ensuring rigorous data quality control.
- Conducted 16-hour multi-modal traffic data surveys and translated empirical sensor evidence into actionable Level of Service (LOS), delay, and queue length analyses via microsimulation (PTV Vissim, SUMO).
- Conducted UAV photogrammetric surveys for 302 schools, 3 resorts, 1 park, and 27 private land developments; processed high-resolution orthophotos and 3D surface models using GIS workflows.

Assistant Urban Planner - Photogrammetry

Sheltech Consultants (Pvt.) Ltd. **November 2024 - April 2025, Dhaka**

- Managed sensor data collection and GIS integration for the feasibility and master planning of 25 towns and 16 Paurashavas (IUGIP Project).
- Processed complex photogrammetric data and integrated drone-based imagery with spatial databases to model urban infrastructure accurately.
- Synthesized high-resolution orthophotos and digital elevation models from UAV imagery for over 400 sq km of urban area, accelerating spatial analysis workflows by 60% and supporting evidence-based decision making for the IUGIP Project using Pix4D, ArcGIS and City Engine.

Urban Planner

Topbright (Pvt.) Ltd **March 2024 - October 2024, Dhaka**

- Directed the spatial data gathering and Master Plan development for the Central Human Resource Development Center.
- Streamlined permit approval workflows by standardizing 12+ zoning compliance documents for the CHRDC site plan review, reducing submission processing time by 30% for municipal stakeholders.
- Conducted stakeholder workshops with 8 municipal officials and planners to gather requirements and align urban development priorities for the CHRDC project, improving consensus-building efficiency by 40%.

GIS Analyst

TORK INC **April 2021 - April 2023, Dhaka**

- Executed large-scale spatial data integration, mapping, and geospatial analysis to support urban infrastructure projects.

- Digitized and processed 17 sq km of aerial drone imagery, enabling high-resolution spatial analysis for two major urban infrastructure projects within project deadlines.
- Utilized QGIS and ArcGIS Desktop to assess the condition and network integration of Institutional infrastructures, delivering actionable reports for two major projects.

RESEARCH PROJECT

AI-Powered Automated Vehicle Counting & Traffic Survey System

Plansculpt Private Consultants • https://youtu.be/P7VO7OKq5FI

- Built a high-performance desktop application integrating PyTorch and TensorRT pipelines for automated, bi-directional vehicle detection and tracking across 18 distinct traffic classes tailored to mixed-traffic conditions.
- Implemented custom directional polygon zones with trajectory logic to accurately capture turning movements, U-turns, and directional flow rates directly from survey video feeds.
- Streamlined traffic survey workflows by developing automated export modules that generate classified count summaries and time-interval slicing directly into Microsoft Excel and CSV formats.

Predictive Modeling of Traffic Impacts using AI Agents Development Project

Plansculpt Private Consultants • https://www.github.com/jahirmiru/SUMO_with_AI

- Engineered an AI-agent-based simulation framework to accurately predict and evaluate the impact of newly generated traffic on existing urban infrastructure.
- Calibrated the predictive algorithms to comply strictly with the Traffic Impact Assessment Guidelines 2025 established by the Dhaka Transport Coordination Authority (DTCA).
- Integrated diverse data sets to simulate mixed-traffic operations, providing actionable, data-driven mitigation strategies.

RESEARCH EXPERIENCE

Exploring Pedestrian Catchment Areas and Factors Influencing Access and Egress Walking Distances of MRT Line-6 Stations in Dhaka

- Collected 2,900 passenger surveys and DAP-based land-use, building-use, road, and footpath data.
- Delineated network-based catchment areas using QNEAT3 in QGIS and conducted GIS-based analysis of land use, building use, roads, and footpaths within 50 m walking-route buffers.
- Applied Multiple Linear Regression (MLR) to assess demographic effects on walking distance and Gradient Boosted Decision Trees (GBDT) to evaluate the relative importance of built-environment variables.

Assessing Highway Condition during Festivals in Bangladesh Class Project

- Conducted multi-modal data integration to evaluate highway infrastructure stress and performance degradation during peak festival periods.
- Analyzed massive traffic volume anomalies and queue lengths using statistical tools to propose scalable predictive interventions.

Suitable Sites for Urban Solid Waste Disposal using GIS Approach in Narayanganj City Corporation

- Utilized Weighted Overlay Analysis and GIS-based multi-criteria decision making to identify optimal locations for infrastructure.
- Produced comprehensive thematic maps and suitability reports for municipal stakeholders.

Accessibility and Safety Audit in BUET Campus Class Project

- Captured and synthesized spatial and observational field data to assess structural safety and pedestrian accessibility across the campus.
- Identified and spatially mapped high-risk pedestrian and structural zones using quantitative safety indices, recommending targeted civil engineering design changes to enhance campus accessibility and resilience.

EDUCATION

Bachelor of Urban and Regional Planning
Bangladesh University of Engineering and Technology (BUET) • Dhaka • 2024 • 2.83
Higher Secondary School Certificate (Science)
Notre Dame College • Dhaka • 2018 • 5.00
•Awarded Government Scholarship
Secondary School Certificate
Shamsul Hoque Khan School and College • Dhaka • 2016 • 5.00

CERTIFICATIONS

Deep Learning Onramp
Mathworks • 2025
Generative AI Leader
Coursera • 2025
Foundations of Project Management
Coursera • 2024
Google Cybersecurity Certification
Coursera • 2023
Fundamental of GIS
Coursera • 2022

INVOLVEMENT

Senior Rover Mate & President
Dhaka • BUET Rover Scout Group • March 2019 - August 2024
•Revitalized the dormant BUET Rover Scout Group after over a decade of inactivity; organized joint national programs with Bangladesh Scouts to restore active campus leadership.
President's Scout Award & Community Development Award
Dhaka • Shamsul Hoque Khan School Scout Group • January 2011 - December 2018
•Conferred the President's Scout Award (highest national youth scout honor) and Community Development Award for distinguished service and leadership.

SKILLS

Transport Modelling: PTV Vissim, Sidra Intersection , SUMO, Synchro with SimTraffic
AI & Machine Learning: Ultralytics YOLO, PyTorch, OpenCV, Automated Object Counting, Deep Learning for Geospatial Data
GIS & Spatial Analysis: QGIS, ArcGIS Desktop, ArcGIS Pro, ArcGIS Online
Programming: Python (GeoPandas, Fiona, Shapely), GEE, MATLAB, R, SQL, Bash, React, Node.js, Django
Sensor & Field Tech: Photogrammetry (OpenDroneMap, Pix4dMapper, Agisoft Metashape), Drone Piloting (DJI pilot, Drone Deploy), RTK GNSS, LiDAR/Spatial Data Processing
Modeling & Infrastructure: Autodesk ReCap Pro, AutoCAD, City Engine, Blender with GIS

REFERENCES

Available upon request